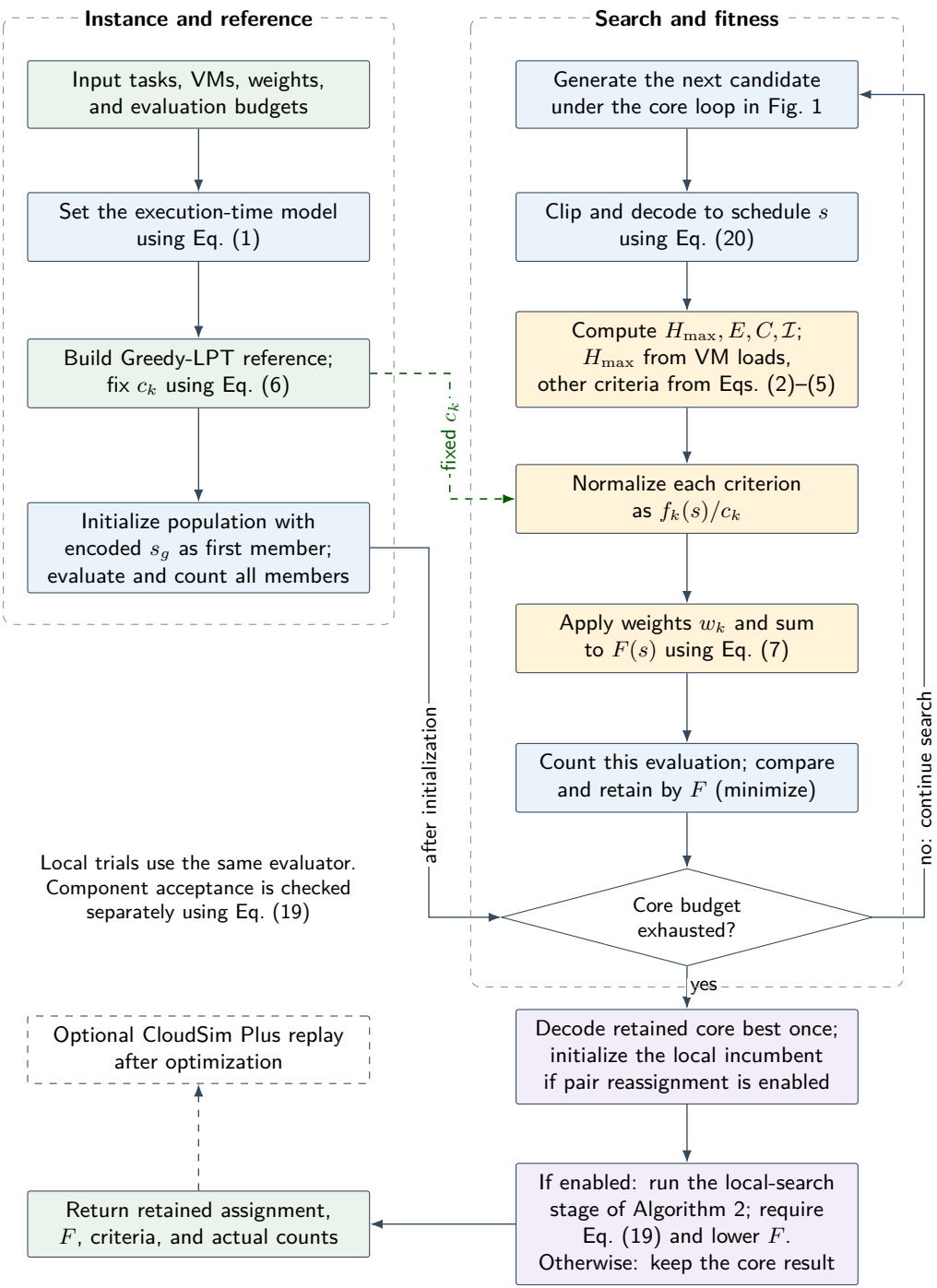

Instance and reference
Search and fitness
Input tasks, VMs, weights, and evaluation budgets
Set the execution-time model using Eq. (1)
Build Greedy-LPT reference; fix c_k using Eq. (6)
Initialize population with encoded s_g as first member; evaluate and count all members
fixed c_k
after initialization
Generate the next candidate under the core loop in Fig. 1
Clip and decode to schedule s using Eq. (20)
Compute $H_{\max}, E, C, \mathcal{I}$; $H_{\max}$ from VM loads, other criteria from Eqs. (2)–(5)
Normalize each criterion as $f_k(s)/c_k$
Apply weights w_k and sum to $F(s)$ using Eq. (7)
Count this evaluation; compare and retain by F (minimize)
Core budget exhausted?
no: continue search
yes
Local trials use the same evaluator. Component acceptance is checked separately using Eq. (19)
Decode retained core best once; initialize the local incumbent if pair reassignment is enabled
If enabled: run the local-search stage of Algorithm 2; require Eq. (19) and lower F. Otherwise: keep the core result
Return retained assignment, F, criteria, and actual counts
Optional CloudSim Plus replay after optimization